Capstone project proposal

Project

Adaptive Wearable Device for Real Time Suppression of Parkinsons Hand Tremor

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Project summary

Parkinson's disease affects roughly ten million people worldwide, and hand tremor is one of its most limiting symptoms. It can make everyday tasks such as eating, writing, and getting dressed much more difficult. Current treatments also have important limitations. Medication can become less effective throughout the day and may cause side effects, while deep brain stimulation can be effective but requires invasive brain surgery. Existing non-invasive wearable devices are mainly designed for essential tremor and often do not work as well for Parkinson's resting tremor. They are also typically calibrated once in a clinical setting and do not adjust as a person's tremor changes due to medication, fatigue, or disease progression. Commercially available options, such as the Cala kIQ neurostimulation wristband, are also cost-prohibitive for many patients, with reported prices starting around $3,200 USD, further limiting access. This project proposes an adaptive wearable device that reduces hand tremor in real time while learning the specific tremor pattern of each user. Motion sensors measure the movement of the hand, and onboard software separates involuntary tremor from intentional movement. Since Parkinson's tremor is generally repetitive, an algorithm can estimate the next part of the tremor before it happens. A semi-active damping actuator then applies a counter-force that opposes the tremor. This predictive approach compensates for actuator delay, which is a major limitation in many active tremor suppression systems. The device is paired with a companion mobile application that continuously records tremor characteristics from the wristband and combines them with user-logged context such as time of day, medication timing, meals, sleep, and daily activities. Over days and weeks, the app builds a longitudinal record of when tremors occur, how severe they are, and what conditions correlate with worsening. This record generates clinician-shareable reports that support medication adjustments and treatment decisions, and it also supplies the labeled data that lets the on-device model keep adapting as the patient's tremor changes with disease progression. The goal is a wearable-and-app system that helps people regain more control of their hands without surgery, with calibration that can adapt over time, and with a record of tremor changes for clinical review.

Engineering scope

The project focuses on five main engineering challenges. The first is a wearable sensing system using an inertial measurement unit and microcontroller to capture three-axis hand motion at a sampling rate sufficient to resolve tremor frequencies and their harmonics. The second is a real-time signal-processing pipeline that separates involuntary tremor from intentional movement using frequency-based filtering and per-axis power analysis. The third is a patient-adaptive prediction model that learns the user's tremor pattern, forecasts motion a short time ahead to compensate for system latency, and periodically re-samples to track disease progression over time. The fourth is a semi-active mechanical actuation system that applies controlled resistance against the predicted tremor while remaining comfortable to wear. The fifth is a companion mobile application that ingests tremor metrics from the wristband over Bluetooth, lets users tag contextual information (time of day, medication, meals, sleep, activity), and generates three views: a real-time status view, a daily and weekly trend view with contextual overlays, and

an exportable clinician-facing report that summarizes symptom patterns and week-over-week change. The app also acts as the labeling mechanism that lets the on-device model continue to adapt as the patient 's tremor drifts. Validation will use public Parkinson's tremor datasets and a mechanical tremor-simulation rig for controlled testing, with performance measured on tremor reduction, system latency, voluntary-motion transparency, and app usability.

Disciplinary contributions

Fouzan and Mazen (Software Engineering) will lead the software stack, including embedded firmware, the real-time signal-processing pipeline, the tremor -prediction model, actuator control logic, and the companion mobile application — spanning BLE data trans port, the local time-series database, the visualization layer, and the clinician-report export. Michael and Imran (Software Engineering, Biomedical Integration) will focus on tremor physiology, sensor selection and placement, ergonomics, contextual-tag design informed by clinical relevance, and clinical evaluation criteria that include both technical metrics (tremor reduction, latency, voluntary-motion transparency) and patient -reported usability and satisfaction metrics. They will also lead validation planning and support the mechanical enclosure and actuator integration, with guidance from a mechanical engineering advisor where needed. Prior team experience with embedded electronics, PCB design, motor control, and 3D -printed enclosures from previous iBioMed design projects supports the hardware side; recent industry experience with sensor development, actuators, and BLE -based logging supports the app-and-firmware pipeline.

Client and stakeholder

A specific clinical stakeholder has not yet been identified. The primary end users are people with Parkinson's disease experiencing hand tremor, while movement-disorder neurologists would be the main clinical stakeholders. A near-term priority is to connect with a movement- disorder clinic for clinical guidance and, if possible, patient feedback through a local hospital, medical school, or Parkinson's support organization. Parkinson Canada has been contacted and a meeting scheduled to discuss the project and secure them as a community stakeholder; outreach is also in progress to the Movement Disorders Clinic at Hamilton Health Sciences via McMaster's Division of Neurology.